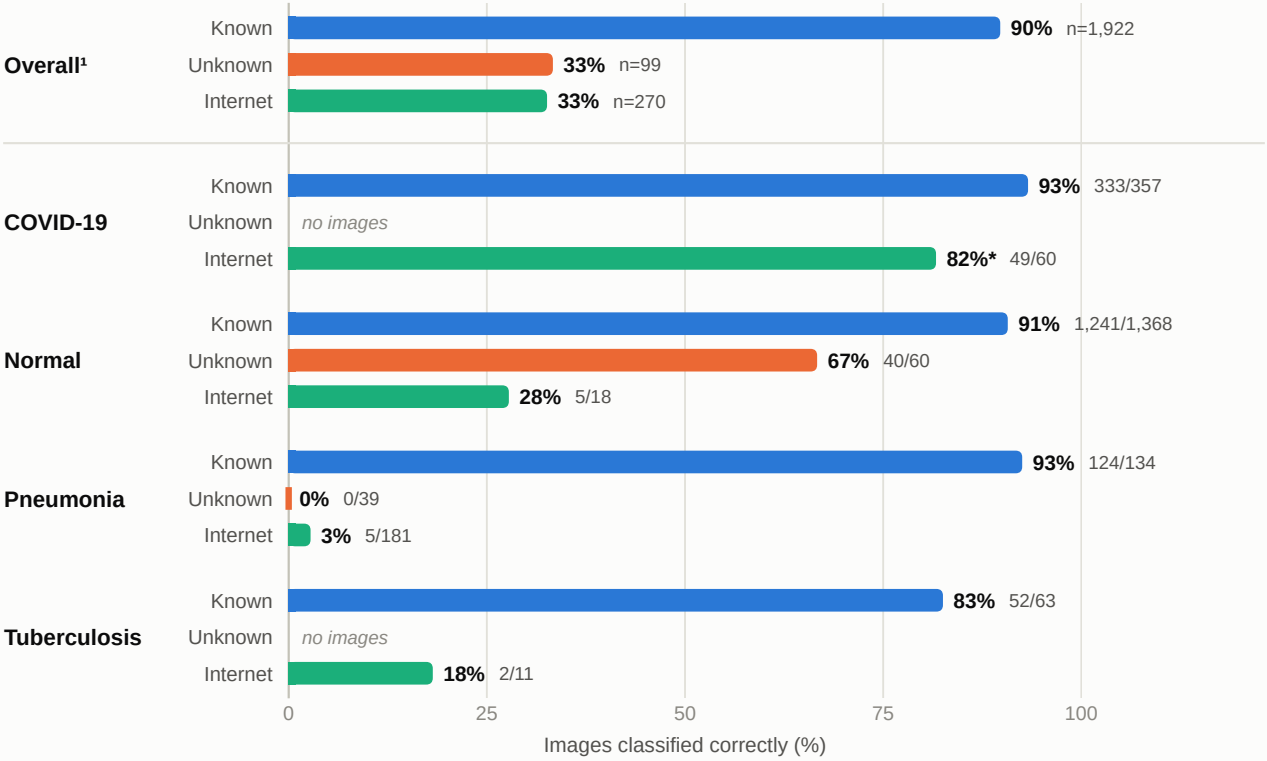
X-ray classifier accuracy on known, unknown and internet images

Share of images the deployed app labelled correctly, by where the images came from

- **Known dataset:** held-out test split of the training data (COVID-19 Radiography + TB Chest X-ray databases)
- **Unknown dataset:** X-rays from a hospital outside the training data (Indiana University; normal and pneumonia only)
- **Internet images:** X-rays gathered from websites and papers (ieee8023 COVID-19 image collection)



Every image was sent through the live app with all checks on; images the app refused count as errors.

¹ Balanced accuracy = mean of the per-class accuracies for the classes in each set, so a set dominated by one class can't look better than it is. Plain accuracy: known 91.1%, unknown 40.4%, internet 22.6%.

* May be optimistic: part of the COVID-19 training data came from the same web collections.